Improvement of our environment: How much green cover is required to reach the optimum of Mumbai city and how can we reach it?

Maryam Ahmad.

Snehal Shirvaiker.

Bhavya Shetty.

Abstract

This study aims to find out how much percentage of green cover Mumbai has compared to the total area of the city and what we can do to make it reach 15 percent of the city. We plan on gathering the needful data from verified sources and reach to a conclusion. Furthermore, do a detailed study on various trees and which ones would be best considering the climate and area constraints. And so, reaching to a final conclusion that would help to better our environment.

Introduction

As we know Oxygen is one of the most essential part of our lives and existence and yet it is neglected the most. There have been many studies regarding deforestation and its adverse effects, yet trees are been cut and urbanization is exponentially increasing. There has also been an increase in so many different illnesses, few which we don’t even know of. This wasn’t the case a few decades back; this being one of the many consequences of urbanization, deforestation and low oxygen levels. If we cannot stop it completely, we should find out some other way to increase the green cover; for the sake of humanity and betterment of the world we live in. Even though there has been an increase in the awareness about this issue, not enough actions have been taken. Even now permits are been passed to cut down thousands of acres of trees but no rules to actually plant them back. Oxygen levels

are dropping each year and that is something everyone should be concerned about. Mumbai is considered as an UHI (Urban Heat Island) i.e. the urban areas have higher temperature than the area with vegetation or a water body in the same city. There have been many influential tree plantation drives done by NGOs like hariyali, HOPE, Isha foundation but that isn't enough to reach the optimum oxygen level that is needed in the environment we live in as of now. So, we, we as in all the people should contribute and coordinate so as to creating a better environment, and this research would be one of those contributions.

Literature reviews

Increasing population and declining biological resources in the context of global change and globalization

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A close relationship between population growth and agricultural land extension, with the latter tied directly to deforestation, has been suggested through a number of cross-country studies notwithstanding fuel wood extraction and logging operations (Billsborrow 1992). While reduction of population growth can reduce local pressures on forests and other natural resources, this relationship is by no means a simple and straightforward one. We in India are now left with a little over 10% of the total land area under closed forest cover. Our forests now are largely confined to upland areas. The most vulnerable ecosystems which are under heavy stress are those in the upland areas – the Himalayas in the north and the north-eastern hills, the Aravalli's in the west, the Vindhya–Sapura ranges in central India and the Eastern and Western Ghats in the south. Growing pressures on these upland ecosystems have serious consequences not only for forest cover and biodiversity, but also for the people in the region and indeed in the sub-continent as a whole. The survival in the lowland plains is wholly dependent upon the health of the mountain ecosystem. At a local level, the worst affected are the most

vulnerable sections of society which include women who have to now work harder to collect their basic needs of fodder and fuelwood.

Cool surfaces and shade trees to reduce energy use and improve air quality in urban areas

Hashem Akbari, Melvin Pomerantz, Haider Taha

Solar energy 70 (3), 295-310, 2001

Elevated summertime temperatures in urban ‘heat islands’ increase cooling-energy use and accelerate the formation of urban smog. Except in the city’s core areas, summer heat islands are created mainly by the lack of vegetation and by the high solar radiation absorptance by urban surfaces. Analysis of temperature trends for the last 100 years in several large U.S. cities indicate that, since ~1940, temperatures in urban areas have increased by about 0.5–3.0°C. Typically, electricity demand in cities increases by 2–4% for each 1°C increase in temperature. Hence, we estimate that 5–10% of the current urban electricity demand is spent to cool buildings just to compensate for the increased 0.5–3.0°C in urban temperatures. Downtown Los Angeles (L.A.), for example, is now 2.5°C warmer than in 1920, leading to an increase in electricity demand of 1500 MW. In L.A., smoggy episodes are absent below about 21°C, but smog becomes unacceptable by 32°C. Because of the heat-island effects, a rise in temperature can have significant impacts. Urban trees and high-albedo surfaces can offset or reverse the heat-island effect. Mitigation of urban heat islands can potentially reduce national energy use in air conditioning by 20% and save over $10B per year in energy use and improvement in urban air quality. The albedo of a city may be increased at minimal cost if high-albedo surfaces are chosen to replace

darker materials during routine maintenance of roofs and roads. Incentive programs, product labelling, and standards could promote the use of high-albedo materials for buildings and roads. Similar incentive-based programs need to be developed for urban trees.

The economic value of trees in urban areas: estimating the benefits of Adelaide's street trees

Phillip Killicoat, Eva Puzio, Randy Stringer

Proceedings Treenet Symposium 94, 106, 2002

As populations become increasingly urbanised, national governments to local councils are recognising that the trees that line our streets, fill our parks and shade our houses make up an urban forest. Urban forests provide multiple benefits that go far beyond adding an aesthetic beauty to our neighbourhoods. Trees in parts, streets and yards, conserve energy, reduce carbon dioxide in the atmosphere, improve air quality, reduce storm runoff, enhance the beauty of our communities by adding colour, texture, and form to our landscapes.

In addition, no matter where trees are located, they represent an interdependent part of complex ecosystems capable of providing a wide range of economic, social and environmental benefits. All these benefits should be considered when attempting to measure the economic, social and environmental benefits of our street trees. These benefits and services, however, are valued differently by different people and different groups in society. Local, regional, national and international interests in our urban trees and the resources they provide also differ greatly and tend to shift over time.

Children living in areas with more street trees have lower prevalence of asthma

G S Lovasi1, J W Quinn1, K M Neckerman1, M S Perzanowski2, A Rundle3

Author affiliations

Abstract

Background: The prevalence of childhood asthma in the USA increased by 50% from 1980 to 2000, with especially high prevalence in poor urban communities.

Methods: Data on the prevalence of asthma among children aged 4–5 years and on hospitalisations for asthma among children less than 15 years old were available for 42 health service catchment areas within New York City. Street tree counts were provided by the New York City Department of Parks and Recreation. The proximity to pollution sources, sociodemographic characteristics and population density for each area were also measured.

Results: Controlling for potential confounders, an increase in tree density of 1 standard deviation (SD, 343 trees/km2) was associated with a lower prevalence of asthma (RR, 0.71 per SD of tree density; 95% CI, 0.64 to 0.79), but not with hospitalisations for asthma (RR, 0.89 per SD of tree density; 95% CI, 0.75 to 1.06).

Conclusions: Street trees were associated with a lower prevalence of early childhood asthma. This study does not permit inference that trees are causally related to asthma at the individual level. The PlaNYC sustainability initiative, which includes a commitment to plant one million trees by the year 2017, offers an opportunity for a large prospective evaluation.

Urban Forestry & Urban Greening

Air pollution removal by trees in public green spaces in Strasbourg city, France☆

Highlights

An application of i-Tree Eco model in Strasbourg city to assess air pollution removal by trees in public green spaces across Strasbourg city was conducted.

The contribution of urban trees in air pollution removal is small in comparison with local emission rates in Strasbourg city.

Public trees of Strasbourg remove about 7% of the emitted PM10coarse in the city's atmosphere.

Abstract

This study integrates i-Tree Eco model in order to estimate air pollution removal by urban trees in Strasbourg city, France. Applied for the first time in a French city, the model shows that public trees, i.e., trees managed by the city, removed about 88 t of pollutants during one year period (from July 2012 to June 2013): about 1 ton for CO; 14 tons for NO2; 56 tons for O3; 12 tons for PM10coarse (particles with diameter ranging from 2.5 to 10 μm); 5 tons for PM2.5 and 1 ton for SO2. Air pollution removal varied mainly with the tree cover and the level of air pollutants concentrations. Comparison between simulated pollution removal rates and local emissions shows that public trees of Strasbourg reduce about 7% of the emitted PM10coarse in the city's atmosphere; however, effect on other air pollutants is small. Thus, our study reveals that urban trees are a significant element to reduce air pollution but are not the only solution to this problem. It's then recommended to associate planting and managing urban forest resources to other strategies that take into account the urban environment characteristics: built structures, street design, location of local sources; etc.

Does forest cover help prevent flood damage? Empirical evidence from India

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Highlights

- The study examines the association between forest cover and flood damage using data from Indian states and flood affected districts of West Bengal.
- Forest cover provides regulatory ecosystem services in the form of protective role against floods.
- Presence of forest cover reduces flood occurrences and damage to human lives and properties.
- Social, economic, demographic and climatic parameters significantly influence the extent of flood damages.
- Both ecosystem-based and traditional hard structural approaches are needed for reducing frequency of flood occurrences and flood related damages.

Abstract

Floods in India are regular phenomenon that occurs in almost all parts of the country, causing significant damage to human lives, assets and ecosystem. Rapid change in weather patterns and decline in forest cover are considered to be the main reasons for frequent floods and associated damage to both human and natural system. Does forest cover help prevent frequent occurrences of flood and reduce flood related damages? Empirical evidences on the flood protective role of forest cover do not give a clear picture. While few studies find a strong link between forest cover and reduced flood incidence, several other studies from across the globe have challenged this traditional notion of the protective role of forest cover. The present study makes an attempt to examine the association between forest cover and flood damage using data from Indian states and the flood affected districts of the eastern Indian state of West Bengal, taking into account the social,

economic, climatic and infrastructural parameters. The findings suggest that forest cover tends to reduce the extent of flood damage and hence has the ability to protect human lives and properties during flood events. It is essential that mitigation and adaptation strategies are shaped in a way that promotes preservation and regeneration of forest resources. Hence, the present study suggests adoption of ecosystem-based adaptation measures along with traditional hard structure flood prevention measures in order to prevent flood related damages effectively.

Analysis of Urban Heat Island (UHI) in Relation to Normalized Difference Vegetation Index (NDVI): A Comparative Study of Delhi and Mumbai

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Abstract

The formation and occurrence of urban heat island (UHI) is a result of rapid urbanization and associated concretization. Due to intensification of heat combined with high pollution levels, urban areas expose humans to unexpected health risks. In this context, the study aims at comparing the UHI in the two largest metropolitan cities of India, *i.e.*, Delhi and Mumbai. The presence of surface UHI is analysed using the Landsat 5 TM image of 5 May 2010 for Delhi and the 17 April 2010 image for Mumbai. The validation of the heat island is done in relation to the Normalized Difference Vegetation Index (NDVI) patterns. The study reveals that built-up and fallow lands record high temperatures, whereas the vegetated areas and water bodies exhibit lower temperatures. Delhi, an inland city, possesses mixed land use and the presence of substantial tree cover along roads; the Delhi Ridge forests and River Yamuna cutting across the city have a high influence in moderating the surface temperatures. The temperature reaches a maximum of 35 °C in West Delhi and a minimum of 24 °C in the east at the River Yamuna. Maximum temperature in East Delhi goes to 30 °C, except the border areas. North, Central and south Delhi have low temperatures (28 °C–31 °C), but the peripheral areas have high temperatures (36 °C–37 °C). The UHI is not very prominent in the case of Delhi. This is proven by the correlations of surface temperature with NDVI. South Delhi, New Delhi and areas close to River Yamuna have high NDVI and, therefore, record

low temperatures. Mumbai, on the other hand, is a coastal city with lower tree cover than Delhi. The Borivali National Park (BNP) is in the midst of dense horizontal and vertical growth of buildings. The UHI is much stronger where the heat is trapped that is, the built-up zones. There are four small rivers in Mumbai, which have low carrying capacity. In Mumbai suburban district, the areas adjoining the creeks, sea and the lakes act as heat sinks. The coastal areas in South Mumbai record temperatures of 28 °C–31 °C; the Bandra-Kurla Complex has a high range of temperature *i.e*., 31 °C–36 °C. The temperature witnessed at Chhatrapati Shivaji International Airport is as high as 38 °C. The temperature is nearly 37 °C–38 °C in the Dorai region in the Mumbai suburban district. The BNP has varied vegetation density, and therefore, the temperature ranges from 27 °C–31 °C. Powai Lake, Tulsi Lake and other water bodies record the lowest temperatures (24 °C–26 °C). There exists a strong negative correlation between NDVI and UHI of Mumbai, owing to less coverage of green and vegetation areas.

Impact of Morphological Characteristics of Green Roofs on Pedestrian Cooling in Subtropical Climates

by **Gaochuan Zhang** [1], **Bao-Jie He** [2,*], **Zongzhou Zhu** [3] and **Bart Julien Dewancker** [1]

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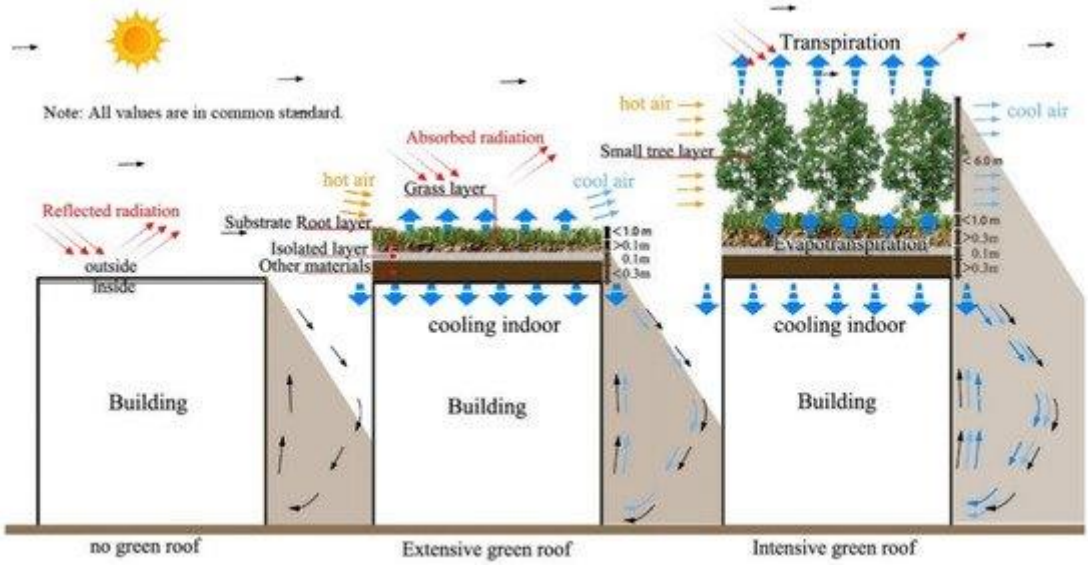
(This article belongs to the Special Issue **Impacts of Urban Overheating on Human Life: the Potential of Mitigation and Adaptation Technologies**)

Abstract

Growing and densifying cities set a challenge for preserving and enhancing green spaces to cool urban spaces. Green roofs, involving the planting of vegetation on rooftops, are regarded as an alternative approach to enhancing urban greenery and urban cooling. For better cooling performances, it is essential to reasonably configure green roofs, especially in real and complex neighbourhoods. Therefore, the aim of this paper is to investigate the impact of morphological characteristics of green roofs on pedestrian cooling in real and complex neighbourhoods. In specific, based on an ENVI-met model, we studied the effect of greening layout, coverage ratio, vegetation height, and building height on pedestrian air temperature reduction in the tropical city of Hangzhou, China. Results indicate green roofs could generate moderate effects on pedestrian air temperature reduction (around 0.10–0.30 °C), while achieving a cooling performance of 0.82 °C. Green roofs in upwind zones were able to generate the most favourable cooling performance, while green roofs in downwind zones

made slight differences to pedestrian thermal environments. Green roofs with a low coverage ratio were not useful for lowering pedestrian temperature, and a greening coverage ratio of 25–75% in upwind zones was cost-effective in real neighbourhoods. Locations that were horizontally close to green roofs enjoyed better cooling performances. Increasing vegetation height could strengthen cooling effects of green roofs, while an increase in building height weakened the cooling performance. Nevertheless, higher building height could enhance pedestrian cooling performances because of building shading effects. In addition, because of wind effects and building shading, building height limits for the cooling performance of green roofs could be higher than 60 m.



Physiological responses of some tree species under roadside automobile pollution stress around city of Haridwar, India

- P. C. Joshi &
- Abhishek Swami

The Environmentalist **volume 27**, pages365–374(2007)Cite this article

- **643** Accesses
- **47** Citations
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Abstract

Studies were carried out to determine the physiological response of few economically important tree species viz., Mango (*Mangifera indica*), *Eucalyptus citriodora*, Sagon (*Tectona grandis*) and Sal (*Shorea robusta*) to roadside automobile pollution during 2004–2005. By determining some physiological parameters, which included chlorophyll *a*, and *b*, total chlorophyll, carotenoids, ascorbic acid, pH and relative water content, impact of automobile exhaust on these species was assessed. The data obtained were further analysed by using one-way ANOVA and a significant change in all these parameters was found in the leaf samples collected from road side trees, exposed to automobile exhausts in comparison to control. Higher value of air pollution tolerance index (APTI) was recorded for *S. robusta* (9.02) while the minimum value of APTI was recorded for *M. indica* (6.76).

Anticipated performance index of some tree species considered for green belt development in an urban area

In the present study, the Air Pollution Tolerance Index (APTI) of ten plant species collected from an urban area has been evaluated by analyzing important biochemical parameters. High values of APTI were recorded in Psidium guajava (31.75%); Swietenia mahoganii(28.08%); Mangifera indica (27.97%); Polyanthia longifolia (25.58%) and Ficus benghalensis (25.02%). The Anticipated Performance Index (API) of these plant species was also calculated by considering their APTI values together with other socio-economic and biological parameters. According to API most tolerant plant species for green belt development were Ficus benghalensis (87%); Mangifera indica (87%); Swietenia mahoganii (87%) and Saraca indica (81%).

Effect of street trees on microclimate and air pollution in a tropical city

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Abstract

One of the fastest growing cities in India, Bangalore is facing challenges of urban microclimate change and increasing levels of air pollution. This paper assesses the impact of street trees in mitigating these issues. At twenty locations in the city, we compare segments of roads with and without trees, assessing the relationship of environmental differences with the presence or absence of street tree cover. Street segments with trees had on average lower temperature, humidity and pollution, with afternoon ambient air temperatures lower by as much as 5.6 °C, road surface temperatures lower by as much as 27.5 °C, and SO_2 levels reduced by as much as 65%. Suspended Particulate Matter (SPM) levels were very high on exposed roads, with 50% of the roads showing levels approaching twice the permissible limits, while 80% of the street segments with trees had SPM levels within prescribed limits. In an era of exacerbated urbanization and climate change, tropical cities such as Bangalore will have to face some of the worst impacts including air pollution and microclimatic alterations. The information generated in this study can help appropriately assess the environmental benefits provided by urban trees, providing useful inputs for urban planners.

Forest cover of a city

https://cdn.narendramodi.in/ebooks/other-ebooks1/tree_cover_in_urban_area/files/assets/basic-html/page22.html

Tree Cover in Urban area - page 22

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4. Tree Cover in Urban Areas in India

The Delhi-based National Institute of Environment Studies (NIES) prepared a list of major cities with tree cover and mentioned that Calcutta has the lowest green cover (less than one per cent of geographical area) among all mega cities[23]. The NIES estimated a tree cover of 8.60 per cent in Bangalore; 11.9 per cent in New Delhi; 6.20 per cent in Greater Mumbai; and 7.5 per cent in Chennai. According to the norms, mentioned by the NIES and some other literature, the tree cover should be at least 15 per cent for megacities with a million-plus population[23]. Thus, majority of the cities, including the green cities, in India have tree cover below the minimum optimum norm. Due to lack of space and demand of the land for other purpose, it may be difficult to achieve the tree cover of 15% of city area or 20 m^2 per inhabitants in the megacities like Ahmedabad and Surat. In such cities, the goal of achieving tree cover of 10 % of the geographical area or 10-15 m^2 tree cover per inhabitant may be a realistic target.

Like the ChineseGovernment, Delhi Government has done commendable job to expand forest and park garden before the Common Wealth Games 2010. At present 20%of Delhi's geographical area is under green cover (11.9%under tree cover and the rest under garden and parks with grasses), making per capita green space availability to around 21.43 m^2. Municipal Corporation of Delhi (MCD), New Delhi Municipal Council (NDMC) and Delhi Development Authority (DDA) have many parks, city forests, biodiversity parks and other green belts. There are nine city forests and two biodiversity parks in Delhi. Nine more city forests are planned to be created. Still there is a need to identify vacant areas which can be put under the green cover. Entire ridge area (about 6,000 ha) needs to be greened.

The status about the tree cover for different cities in India is not available, but the Forest Survey of India (2011)[20] reported forest and tree cover in the two Urban Areas (Chandigadh and Delhi) of the Union Territories. The Gujarat Forest Department conducted a separate study (2011)[21] to assess tree cover in Gandhinagar Town Planning Area, and the comparable figures have been given in Table-1.

InGujarat, about 26.9 crore trees (TOF) have tree cover of 7837 sq. km., i. e 343 trees equivalent to one hectare of tree cover. A study based on remote sensing data has estimated tree cover over an area of 3,075 ha in Gandhinagar which is equivalent to 8.67 lakh trees (both estimates are of the year 2011). Thus, about 282 trees, distributed in the different girth classes above 10 cm, may be equivalent to one hectare of tree cover in Gujarat. Since trees having lower girth classes are relatively less and majority belong to medium and large size trees, this equation appears logical. The calculations regarding tree cover in the cities presented in this report are based on this equation. Sacramento's urban forest USA) has 13% tree cover which is equivalent to 73 trees/ha[26].

THE VALUE OF GREEN INFRASTRUCTURE FOR URBAN CLIMATE ADAPTATION

The Center for Clean Air Policy

(About CCAP Since 1985, CCAP has been a recognized world leader in climate and air quality policy and is the only independent, non-profit think-tank working exclusively on those issues at the local, national and international levels. Headquartered in Washington, D.C., CCAP helps policymakers around the world to develop, promote and implement innovative, market-based solutions to major climate, air quality and energy problems that balance both environmental and economic interests. For more information about CCAP, please visit www.ccap.org.)

February 2011

Josh Foster, Ashley Lowe, Steve Winkelman

“Urban systems provide ideal laboratories for understanding resiliency and for developing dual-use technologies, practices, and systems that provide value even if no negative events occur.”1 Over the coming century, climate change scenarios project that urban regions will be expected to manage extremes of precipitation and temperature, increased storm frequency and intensity, and sea-level rise. Increases in problems with which urban areas are already coping may be indicating– or at least mimicking – that climate change impacts are already occurring and are likely to worsen in the future.2 In practice, these impacts will mean coping with: • Longer and hotter heat waves • Increased urban heat island (UHI) impacts such as heat related illness and higher cooling demand and costs • More damaging storms and storm surges • Greater river flooding • Increased frequency and intensity of combined sewer overflows (CSOs) • More intense and extended duration of droughts • Longer water supply shortages, and • Declines in local ecosystem services, such as the loss of coastal wetlands that buffer communities against hurricanes.

Applications of these green infrastructure approaches range in scale from individual buildings, lots, and neighborhoods to entire cities and metro regions. Green infrastructure strategies can be implemented via large, centralized public “macro” projects or through smaller, decentralized “micro” applications on private property.12

Therefore, the benefits of green infrastructure can be measured at the building or site level such that the individual owners reap the private benefits or, if spread across many private owners, the benefits can be aggregated to an entire community, city, county, region, or even nation.

However, to achieve these benefits of scale there must be coordinated implementation across a broader area involving multiple parties to reach certain critical levels of participation. Consequently, community-level, rather than individual-level implementation of green infrastructure particularly helps local governments to achieve environmental, sustainability, and adaptation goals within their jurisdictions. The climate adaptation benefits of green infrastructure are generally related to its ability to moderate the expected increases in extreme precipitation or temperature. Benefits include better management of storm-water runoff, lowering incidents of combined storm and sewer overflows (CSOs), water capture and conservation, flood prevention, accommodation of natural hazards (e.g., relocating out of floodplains), reduced ambient temperatures and urban heat island (UHI) effects, and defense against sea level rise (with potential of storm-surge protection measures). The U.S. Environmental Protection Agency (EPA) has also identified green infrastructure as a contributor to improving human health and air quality, lowering energy demand, reducing capital cost savings, increasing carbon storage, expanding wildlife habitat and recreational space, and even increasing land-values by up to 30%.13 Given the above benefits, green infrastructure approaches help to achieve sustainability and resilience goals over a range of outcomes, in addition to climate adaptation. The value of green infrastructure is calculated by comparing the costs of green practices to “hard” infrastructure alternatives, the value of avoided damages, or market preferences that enhance value, like property value.14 Green infrastructure benefits generally can be divided into five categories of environmental protection: (1) Land-value, (2) Quality of life, (3) Public health, (4) Hazard mitigation, and (5) Regulatory compliance.15 Green technologies and infrastructure solutions are often implemented with a single goal in mind, such as managing storm-water or reducing local ambient heat, and the costs and benefits are often evaluated in the same way. However, their full net-benefit can only be realized by a comprehensive accounting of their multiple benefits. For example, trees filter water, slow runoff, cool local and regional urban heat effects, and clean air.

Eco-Roofs

In terms of climate adaptation, eco-roofs are generally installed to respond to two primary climate drivers–extreme precipitation and

temperature. There are three main types of eco-roofs: Green roofs (vegetated), white roofs (cooling), and blue roofs (water managing). Green, blue, and white roofs have distinct and overlapping benefits compared to typical “black” roofs meant solely to provide shelter. Communities or building owners with limited budgets, who are primarily interested in energy savings or reducing peak energy demand, generally focus on cool roofs. Those who can consider life-cycle costs and public benefits, and who are interested in broader environmental impacts (particularly improving storm-water management) may choose to install green roofs. Sustainability leaders, such as Chicago and New York City, recognize the value and opportunity for both cool and green roof technologies and are supporting efforts to encourage both options.16 Eco-roofs are usually established to achieve additional environmental and sustainability goals, including: • Water conservation • Storm-water runoff and water quality management • Local and regional cooling • Aesthetic value • Electricity savings • Habitat provision for wildlife • Carbon absorption There are three primary types of eco-roofs: Green roofs, White roofs and blue roofs. The next sections will describe the properties, costs and benefits associated with each type of eco-roof.

White Roofs–

Adapting to the Urban Heat Island Effect The urban heat island effect is caused by the tendency of hard, dark surfaces, such as roofs and pavement, to be measurably hotter than natural areas. It can raise a city’s temperatures 2 to 5.5° C on hot summer days. White or cool roofs are generally flat roofs that have been painted white or are surfaced with some other light or reflective material—often adding durability while reducing ambient temperatures. Research reveals that conventional roofs can be 31-55° C hotter than the air on any given day, while cool roofs tend to stay within 6-11°C of the background temperature. This cooling performance can lower ambient temperature, mitigate the UHI, and help prevent mortality during heat waves.33 White vinyl roofs are the most reflective common material used, reflecting 80% of the sun’s rays compared to only 6% reflection on a conventional black roof and avoiding 70% of the heat absorption experienced on black roofs. Some coatings can reach even higher levels of reflectivity.

Urban Forestry

Planting and maintaining trees in urban settings is considered a quintessential green infrastructure practice with multiple benefits for

resilience, adaptation, and even climate mitigation. The benefits of urban forestry extend from individual neighborhood trees to widely distributed urban forests. As noted earlier, trees contribute to adaptation by intercepting and filtering storm-water runoff to prevent flooding and improve water quality, absorbing pollutants to clean the air, providing wind-breaks to protect buildings from wind damage, and regulating heat island effects through shading and evaporation. Simultaneously, trees contribute to mitigation by lowering cooling demand for electricity and directly sequestering carbon. Trees provide wildlife habitat and ecosystem services, and have been shown to increase property values. For centuries they have contributed to overall urban quality of life. Dead trees even can be recycled into mulch. Urban forestry programs establish trees in public spaces such as parks, along streets and alleys, or in any available open areas that local governments manage (along stream rightof-ways, around public buildings, or in city-owned vacant lots). Urban forestry can extend to green-belts around cities that buffer waterways and regulate development, and even to acquisition and management of lands to preserve urban watersheds so that drinking water supply and quality is protected. Local ordinances often guide property owners' responsibilities for trees as a private and public good. Urban forestry delivers a range of stormwater and UHI benefits to communities. A typical medium-sized tree can intercept as much as 2,380 gallons of rainfall per year.72 During the summer, with trees in full leaf, evergreens and conifers in Sacramento, CA were found to intercept over 35% of the rainfall that hit them during smaller rainfall events. Trees reduce runoff and erosion from storms by about 7% and reduce the need for erosion control.73 In Oakland, CA the continuous tree canopy is estimated to intercept 4 inches of rain, or 108,000 gallons of water, per acre in a typical year. 74 Trees can reduce runoff in urban areas by up to 17%.75 In terms of mitigating UHI impacts, trees typically absorb 70 to 90% of sunlight in summer and 20 to 90% in winter, with the biggest seasonal variation seen in deciduous trees (versus evergreens) that lose their leaves annually. One study showed that trees can reduce the maximum surface temperature of the roofs and walls of buildings by 11 to 25º C. The estimated effect of new shade trees planted around houses resulted in annual cooling energy savings of 1% per tree while annual heating energy use decreased by almost 2% per tree.76 Direct energy savings from shading by trees and vegetation could reduce carbon emissions in various U.S. metropolitan areas by roughly 1.5 to 5% due to decreases in cooling energy use.77 A climate modeling study

in Manchester, England found that adding 10% green cover like grass and shrubs in high density areas would keep surface temperatures below historical baseline levels, except under conditions of high CO_2 emissions. In the model, when green roofs were also added to high density areas, temperatures stayed below baseline levels even under the high emissions scenarios. Conversely, the studies also show that if green cover is left unaltered, temperatures are expected to increase about 3.3 to 3.8°C.78 In addition to these benefits, trees absorb and reduce various pollutants found in the urban environment, including particulate matter (PM), nitrogen oxides (NO_X), sulfur dioxide (SO_2), carbon monoxide (CO), and ground-level ozone (O_3). One study predicted that increasing the urban canopy of New York City by 10% could lower ground-level ozone by about 3%. Another study estimated that one million additional trees in a city could lower emissions of NO_x by almost a quarter ton per day and particulate matter by over one ton per day. A 2006 study estimated that urban trees in the United States remove 784,000 tons of pollutant per year at an economic value of $3.8 billion. The study focused only on deposition of ground-level ozone, PM, NO_2 , SO_2 and CO. Although the estimated changes in local ambient air quality were modest, typically less than 1%, the study noted that additional benefits would be gained if urban temperature and energy impacts from trees and vegetation were also included.

Deforestation is leading to more infectious diseases in humans

As more and more forest is cleared around the world, scientists fear that the next deadly pandemic could emerge from what lives within them.

A direct link

Malaria—which kills over a million annually due to infection by *Plasmodium* parasites transmitted by mosquitoes—has long been suspected of going hand in hand with deforestation. In Brazil, while control efforts have dramatically reduced malaria transmission in the

past—bringing 6 million cases a year in the 1940s down to just 50,000 by the 1960s—cases have since been steadily rising again in parallel with rapid forest clearing and expansion of agriculture. At the turn of the century, there were over 600,000 cases a year in the Amazon basin.

Work in the late 1990s by Amy Vittor, an epidemiologist at the University of Florida's Emerging Pathogens Institute, and others, suggested a reason why. Clearing patches of forest appears to create ideal habitat along forest edges for the mosquito *Anopheles darlingi*—the most important transmitter of malaria in the Amazon—to breed. Through careful surveys in the Peruvian Amazon, she found higher numbers of larvae in warm, partially shaded pools, the kind that form beside roads cut into forests and puddles behind debris where water is no longer taken up by trees.

In a complex analysis of satellite and health data published recently in the journal *Proceedings of the National Academy of Sciences,* MacDonald and Stanford University's Erin Mordecai reported a significant impact of deforestation across the Amazon basin on malaria transmission, in line with some previous research.

Between 2003 and 2015, on average, they estimated that a 10 percent yearly increase in forest loss led to a 3 percent rise in malaria cases. For example, in one year of the study, an additional 618-square-mile (1,600-square-kilometer) patch of cleared forest—the equivalent of nearly 300,000 football fields—was linked to an additional 10,000 cases of malaria. This effect was most pronounced in the interior of the forest, where some patches of forest are still intact, providing the moist edge habitat that the mosquitoes like.

With the ongoing burning of the Amazon, these results don't bode well; The latest data, issued this week, reveals an area 12 times the size of New York City has been destroyed so far this year.

"I am concerned about what's going to happen with transmission following the end of the fires," MacDonald says.

It's hard to generalize about mosquito ecology, which varies depending on species and region, Vittor stresses. In Africa, studies have found little association between malaria and deforestation—perhaps because the mosquito species there like to breed in sunlit bodies of water and favor open farmland over shady forest areas. But in Sabah, a part of Malaysian Borneo, malaria outbreaks also occur in tandem with bursts of forest clearing for palm oil and other plantations.

Fever from the jungle

Mosquitoes aren't the only animals that can transmit deadly scourges to people. In fact, 60 percent of new infectious diseases that emerge in people—including HIV, Ebola, and Nipah, all of which originated in forest-dwelling animals—are transmitted by a range of other animals, the vast majority of them wildlife.

In a 2015 study, researchers at Ecohealth Alliance, a New York-based non-profit that tracks infectious diseases globally, and others found that "nearly one in three outbreaks of new and emerging disease[s] are linked to land-use change like deforestation," the organization's president Peter Daszak tweeted earlier this year.

Many viruses exist harmlessly with their host animals in forests, because the animals have co-evolved with them. But humans can become unwitting hosts for pathogens when they venture into or change forest habitat.

"We are completely changing the structure of the forest," notes Carlos Zambrana-Torrelio, a disease ecologist at Ecohealth Alliance.

Deadly attraction

Diseases can also occur when new habitats draw disease-carrying species out of the forest.

For instance, in Liberia forest clearings for palm oil plantations attract hordes of typically forest-dwelling mice, lured there by the abundance of palm fruit around plantations and settlements. Humans can contract Lassa virus when they come into contact with food or objects contaminated with feces or urine of virus-carrying rodents or bodily fluids of infected people. In humans, the virus causes hemorrhagic

fever—the same kind of illness triggered by Ebola virus—and in Liberia killed 36 percent of infected people.

Virus-carrying rodents have also been spotted in deforested areas in Panama, Bolivia, and in Brazil. Alfonso Rodriguez-Morales, a medical researcher and tropical disease expert at Colombia's Universidad Tecnológica de Pereira, fears that their ranges will increase following the resurgence of fires in the Amazon this year.

Such processes aren't limited to tropical diseases. Some of MacDonald's research has revealed a curious association between deforestation and Lyme disease in the Northeastern United States.

Borrelia burgdorferi, the bacterium that causes Lyme disease—is transmitted by ticks that rely on forest-dwelling deer to breed and obtain enough blood to survive. However, *the bacterium is also found in the* white-footed mouse, which happens to thrive in forests fragmented by human settlements, MacDonald says.

Spillovers of infectious diseases to people is more likely to occur in the tropics because overall wildlife and pathogen diversity is higher, he adds. There, a number of diseases transmitted by a wide range of animals—from blood-sucking bugs to snails—have been linked to deforestation. On top of known diseases, scientists fear that a number of yet-unknown deadly diseases are lurking in forests that could be exposed as people encroach further.

Zambrana-Torrelio notes that the likelihood of spillovers to people may increase as the climate warms, pushing animals, along with the viruses they carry, into regions where they've never existed before, he says.

Whether such diseases stay confined to forest fringes or if they gain their own foothold in people, unleashing a potential pandemic, depends on their transmission, Vittor says. Some viruses, like Ebola or Nipah, can be transmitted directly between people, theoretically allowing them to travel around the world as long as there are humans.

Zika virus, which was discovered in Ugandan forests in the 20th century, could only cruise the world and infect millions because it found a host in *Aedes aegpti*, a mosquito that thrives in urban areas.

“I’d hate to think that another or several other pathogens could do such a thing, but it’d be foolish not to think of that as a possibility to prepare for,” says Vittor.

A new service

Ecohealth Alliance researchers have proposed that containing diseases could be considered a new ecosystem service, that is, a benefit that humans freely gain from natural ecosystems, just like carbon storage and pollination.

To make that case, their team has been working in Malaysian Borneo to itemize the exact cost of malaria, down to each hospital bed, and syringe that doctors use. On average, they found that the Malaysian government spends around $5,000 to treat each new malaria patient in the region—in some areas much more than they spend on malaria control, Zambrana-Torrelio says.

Over time, that adds up, outweighing the profits that could be gained by cutting forests down and making a compelling financial argument to leave some forests standing, Daszak says.

He and his colleagues are beginning work with the Malaysian government to incorporate this into land use planning, and are undertaking a similar project with Liberian officials to calculate the cost of Lassa fever outbreaks there.

MacDonald sees value in this idea: “If we can conserve the environment, then perhaps we can also protect health,” he says. “That I think is the silver lining that we should keep in mind.”

References:

Google Scholar